Advanced Functions

ANSWER KEY

Section 1: Exponential Functions & Growth/Decay

1 A bacterial culture doubles in population every 4 hours. If the initial count is 500, what is the population after 14 hours?

a. $4,000$

b. $4,000\sqrt{2}$ ✓

c. $8,000$

d. $16,000$

Population formula: $P(t) = 500 \times 2^{t/4}$. For $t = 14$:
$P(14) = 500 \times 2^{3.5} = 500 \times 2^3 \times 2^{0.5} = 500 \times 8 \times \sqrt{2} = 4,000\sqrt{2}$.

2 An investment of $2,000$ earns 6% annual interest compounded continuously. What is the value after t years?

a. $2000(1.06)^t$

b. $2000e^{0.06t}$ ✓

c. $2000(1 + \frac{0.06}{12})^{12t}$

d. $2000e^{6t}$

Continuous compounding uses the formula $A = Pe^{rt}$. Substituting $P = 2000$ and $r = 0.06$ gives $A = 2000e^{0.06t}$.

3 Solve the exponential equation for x: $4^{x+1} = 8^{2x-1}$.

a. $\frac{1}{2}$

b. $\frac{5}{4}$ ✓

c. $\frac{3}{2}$

d. 2

Express with base 2: $(2^2)^{x+1} = (2^3)^{2x-1} \Rightarrow 2^{2x+2} = 2^{6x-3}$. Equate exponents:
$2x + 2 = 6x - 3 \Rightarrow 4x = 5 \Rightarrow x = \frac{5}{4}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Logarithmic Laws & Equations

4 Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. 4 ✓

b. 2

c. -2

d. 4 and -2

Combine logs: $\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since $\log_2(x)$ requires $x > 0$, the only valid solution is $x = 4$.

5 Which of the following is equivalent to $\frac{\ln 25}{\ln 5}$?

a. $\ln 5$

b. $\ln 20$

c. 2 ✓

d. 5

Rewrite numerator: $\ln 25 = \ln(5^2) = 2\ln 5$. Therefore, $\frac{2\ln 5}{\ln 5} = 2$.

6 If $\log_b 3 = x$ and $\log_b 5 = y$, what is $\log_b(45)$ in terms of x and y?

a. $2x + y$ ✓

b. $x^2 + y$

c. $2xy$

d. $x + 2y$

Factor $45 = 3^2 \times 5$. Using log properties: $\log_b(3^2 \times 5) = 2\log_b 3 + \log_b 5 = 2x + y$.

Section 3: Rational Functions & Asymptotes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 What is the slant (oblique) asymptote of the rational function $f(x) = \frac{2x^2 + 5x - 3}{x + 1}$?

a. $y = 2x + 3$ ✓

b. $y = 2x - 1$

c. $y = 2x + 5$

d. $y = 2x$

Perform polynomial division: $\frac{2x^2 + 5x - 3}{x + 1} = (2x + 3) - \frac{6}{x + 1}$. As $x \to \pm\infty$, the remainder approaches zero, giving slant asymptote $y = 2x + 3$.

8 Where does the rational function $g(x) = \frac{x^2 - 4}{x^2 - x - 2}$ have a removable discontinuity (hole)?

a. $x = 2$ ✓

b. $x = -1$

c. $x = -2$

d. No holes

Factor: $\frac{(x-2)(x+2)}{(x-2)(x+1)}$. The common factor $(x - 2)$ cancels, indicating a hole at $x = 2$ and a vertical asymptote at $x = -1$.

9 What is the domain of $h(x) = \ln(x^2 - 9)$?

a. $(-3, 3)$

b. $(-\infty, -3) \cup (3, \infty)$ ✓

c. $[3, \infty)$

d. $x \neq \pm 3$

The argument of natural log must be strictly positive: $x^2 - 9 > 0 \Rightarrow (x - 3)(x + 3) > 0$. This holds for $x < -3$ or $x > 3$.

Section 4: Parametric & Polar Curves

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Eliminate the parameter t in the parametric equations $x = 3\cos t$ and $y = 3\sin t$ to find the Cartesian equation.

a. $x^2 + y^2 = 9$ ✓

b. $x^2 - y^2 = 9$

c. $y = 3x$

d. $x + y = 3$

Square and add: $x^2 + y^2 = (3\cos t)^2 + (3\sin t)^2 = 9(\cos^2 t + \sin^2 t) = 9$.

11 A curve is defined by $x(t) = 2t - 1$ and $y(t) = t^2 + 3$. What is y expressed directly as a function of x?

a. $y = \frac{(x+1)^2}{4} + 3$ ✓

b. $y = (2x - 1)^2 + 3$

c. $y = \frac{x^2+1}{2} + 3$

d. $y = \frac{(x-1)^2}{4} + 3$

From $x = 2t - 1$, solve for t: $t = \frac{x+1}{2}$. Substitute into y: $y = (\frac{x+1}{2})^2 + 3 = \frac{(x+1)^2}{4} + 3$.

12 Convert the polar equation $r = 4\cos\theta$ into an equivalent Cartesian equation.

a. $(x - 2)^2 + y^2 = 4$ ✓

b. $x^2 + (y - 2)^2 = 4$

c. $x^2 + y^2 = 4$

d. $(x - 4)^2 + y^2 = 16$

Multiply both sides by r: $r^2 = 4r\cos\theta$. Substitute $r^2 = x^2 + y^2$ and $x = r\cos\theta$:
$x^2 + y^2 = 4x \Rightarrow x^2 - 4x + 4 + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647